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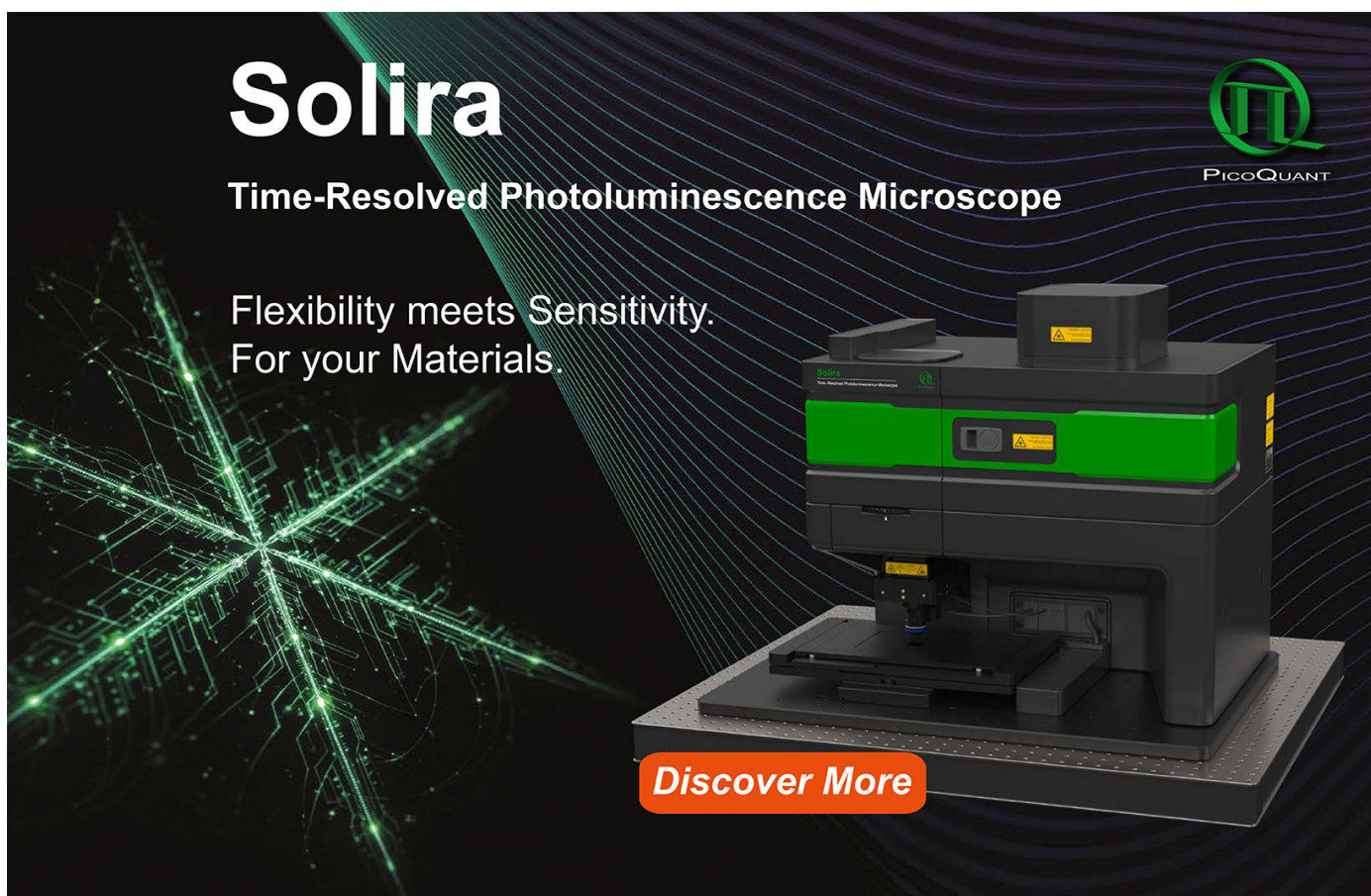
ECG beat classification based on dilated multi-scale ResNet and BiLSTM

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ECG beat classification based on dilated multi-scale ResNet and BiLSTM

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Abstract

Cardiovascular diseases remain a major cause of death worldwide, and electrocardiogram (ECG) analysis is an important tool for heartbeat abnormality detection. However, existing intelligent ECG beat classification methods often fail to adequately capture complex waveform morphology and temporal dependency, or they require a large number of parameters, which limits deployment efficiency. In this study, we propose an ECG beat classification framework that combines a dilated multi-scale residual network (ResNet) with a bidirectional long short-term memory network (BiLSTM). Main experiments, ablation studies, and patient-wise internal evaluation were conducted on the MIT-BIH Arrhythmia Database, and an additional external validation was performed on the INCART Arrhythmia Database. For the MIT-BIH dataset, a patient-wise split was adopted to ensure no patient overlap among the training, validation, and test sets, and SMOTE was applied only to the training set after dataset splitting. The ECG recordings were preprocessed through baseline-drift suppression, wavelet denoising, and heartbeat segmentation. A multi-scale module was introduced into the ResNet backbone to extract multi-scale morphological features from ECG beats, and a BiLSTM layer was incorporated to model temporal dependencies within the extracted feature sequence of each heartbeat segment. Furthermore, standard convolutions were replaced with dilated convolutions to enlarge the receptive field while controlling parameter growth. Component-wise ablation experiments on the MIT-BIH dataset verified the effectiveness of the multi-scale module, BiLSTM layer, and dilated convolution. On the MIT-BIH Arrhythmia Database, the proposed method achieved an overall classification accuracy of 99.39%. In external validation on the INCART database, the model achieved an overall accuracy of 98.75%, suggesting promising cross-dataset generalization potential. These results indicate that the proposed framework can effectively characterize multi-scale ECG beat morphology and temporal feature dependencies while maintaining favorable model efficiency. Nevertheless, further validation on larger multi-center clinical datasets is still required before broader clinical application.

1. Introduction

Cardiovascular diseases (CVDs) are associated with multiple risk factors, including population aging, unhealthy lifestyles, metabolic disorders, hypertension, and genetic susceptibility. They remain a major global public health burden and one of the leading causes of mortality worldwide [1]. Electrocardiogram (ECG) monitoring is a non-invasive, low-cost, and widely used diagnostic tool for cardiac rhythm assessment, and it plays an important role in arrhythmia detection and clinical decision support.

Arrhythmias, which account for a major proportion of CVD, are traditionally diagnosed by doctors' manual examination of ECG [2, 3]. However, the increasing number of CVD patients has resulted in a growing workload for doctors, leading to a higher incidence of 'misdiagnosis and missed diagnosis' in manual analysis. With advancements in artificial intelligence and machine learning (ML), ECG-based

intelligent analysis methods have been explored for the auxiliary diagnosis of arrhythmias, demonstrating promising potential in experimental and clinical studies. The intelligent methods for the auxiliary diagnosis mainly include ML methods [4] and deep learning (DL) methods [5]. ML-based methods analyze ECG signals using artificially extracted arrhythmia features, thereby diagnosing the condition. For instance, Li *et al* [6] initially extracted the corresponding approximate entropy as handcrafted features based on the wavelet packet decomposition algorithm, and combined it with the support vector machine (SVM) algorithm to diagnose arrhythmias, achieving an average accuracy rate of over 97%. Li [7] utilized a method of manually designed features for the detection of five types of arrhythmias. The designed features including wavelet features and time-domain features related to arrhythmia diagnosis, with an average accuracy rate exceeding 91%. Baek *et al* [8] analyzed 2412 ECG image datasets and developed a ML algorithm based on ECG for identifying atrial fibrillation during sinus rhythm, with the predictive model achieving an area under the curve (AUC) value of 0.75 in external validation. Chang *et al* [9] analyzed 2515 ECG datasets using a convolutional neural network algorithm to identify ventricular premature beats during normal sinus rhythm, constructing a single-input image model and a multi-input time series model, with accuracies reaching 89.5% and 88.0%, respectively. Sangha *et al* [10] further developed a diagnostic model based on ML that includes at least six different rhythm abnormalities or conduction defects, using 22 28 236 ECG images. The AUC value of the ML-ECG model in external validation was 0.9.

Although ML-based auxiliary diagnostic methods for arrhythmia have shown promising results, they often rely heavily on manual feature extraction in the preliminary phase. Certain ECG beat morphology patterns may be difficult to fully characterize, potentially impacting the model's generalization ability and recognition accuracy, particularly in diverse clinical scenarios.

DL-based ECG beat classification methods integrate deep representations of ECG waveforms with beat-level classification. After training with a large amount of ECG data, they can automatically learn discriminative features of different heartbeat classes for classification. Recent studies have proposed various DL-based ECG classification methods. For example, Yildirim *et al* [11] utilized a bidirectional long short-term memory (BiLSTM)-wavelet model for classifying five types of ECG signals, achieving an accuracy of 99.39%. Hou *et al* [12] developed an LSTM-based auto-encoder model for ECG arrhythmia classification. Chen *et al* [13] combined CNN and LSTM networks to classify arrhythmia signals using the MIT-BIH database. Wang *et al* [14] proposed a convolutional neural network with a non-local convolutional attention module (NCBAM) for the purpose of automatic classification of ECG rhythms, obtaining an F1 average score of 0.9664 on the MIT-BIH Arrhythmia Database. Yan *et al* [15] proposed a patient-specific ECG classification method using CNNs and spiking neural networks, accomplishing a five-category classification of the MIT-BIH ECG database. Acharya *et al* [16] employed CNNs to automatically detect normal and myocardial infarction ECG signals under both noisy and noise-free conditions, reaching accuracies of 93.5% and 95.2%, respectively.

Although existing ECG beat classification methods have achieved promising performance, several limitations remain. First, conventional ML methods rely heavily on handcrafted features, which may be insufficient to fully characterize the complex and diverse morphology of ECG beats, thereby limiting generalization in different signal conditions. Second, many CNN-based methods mainly focus on local morphological feature extraction with fixed receptive fields, which may not adequately capture the multi-scale waveform variations of different heartbeat categories, especially when narrow and wide abnormal beats coexist. Third, although recurrent architectures such as LSTM can model temporal dependency, existing hybrid CNN-LSTM methods often increase model complexity and parameter quantity, which is unfavorable for efficient deployment. In addition, some studies train and test on ECG segments with very short temporal intervals or highly similar signal contexts, which may lead to optimistic performance estimates and reduced robustness in practical applications.

Recent ECG beat classification studies have mainly focused on three directions: enhancing local morphological representation through improved convolutional modules, modeling temporal dependency by recurrent or attention-based structures, and reducing model complexity for efficient deployment. However, many existing methods still face one or more limitations, such as insufficient minority-class recognition, limited use of bidirectional rhythm context, or increased parameter growth after introducing hybrid architectures. These observations motivated the present study to design a framework that jointly considers multi-scale morphology modeling, temporal dependency learning, and parameter-efficiency control.

To address these limitations, we propose an ECG beat classification framework based on a dilated multi-scale residual network (ResNet) and BiLSTM. ResNet is selected as the backbone because its residual architecture provides stable optimization, hierarchical morphological feature extraction, and a convenient basis for integrating the proposed multi-scale and dilated modules. Specifically, a multi-scale

residual structure is introduced to better capture ECG waveform morphology at different temporal scales, a BiLSTM layer is incorporated to model temporal dependencies within the extracted heartbeat feature sequence, and dilated convolution is employed to enlarge the receptive field while controlling parameter growth. For internal model development and evaluation, the MIT-BIH Arrhythmia Database was split using a patient-wise scheme (60% training, 20% validation, and 20% test) to ensure no patient overlap between sets. In addition, denoising, heartbeat segmentation, and training-set class balancing using SMOTE after patient-wise splitting were included in the preprocessing pipeline to improve data quality and mitigate class imbalance. To further examine cross-dataset generalization, an additional external validation experiment was conducted on the INCART Arrhythmia Database without fine-tuning. Through this design, the proposed framework is expected to achieve a better balance among morphological representation, temporal modeling, model efficiency, and generalization assessment.

The main contributions of this paper are outlined as follows:

- 1) A dilated multi-scale residual module is designed to extract ECG morphological features at different temporal scales. By combining convolutional branches with different kernel sizes and dilated convolutions, the proposed module captures both fine local transitions and broader waveform configurations without substantially increasing the number of learnable parameters.
- 2) A BiLSTM layer is integrated after the multi-scale convolutional representation to model temporal dependencies within the extracted heartbeat feature sequence. This design enables the network to jointly exploit local morphology, multi-scale waveform structure, and sequential feature information for ECG beat classification.
- 3) A patient-wise evaluation protocol is adopted on the MIT-BIH Arrhythmia Database to avoid patient overlap among the training, validation, and test sets. SMOTE is applied only to the training set after patient-level splitting, while the validation and test sets remain unchanged, thereby reducing the risk of data leakage caused by synthetic samples.
- 4) In addition to internal evaluation and ablation studies on the MIT-BIH database, an external validation experiment is conducted on the INCART Arrhythmia Database without fine-tuning or domain adaptation, providing a more transparent assessment of cross-dataset generalization.

2. Dataset description and related work

This section provides an overview of the datasets used in this study and summarizes related studies on ECG beat classification.

2.1. Introduction to the dataset

The MIT-BIH Arrhythmia Database was used as the primary dataset for model development, ablation studies, and internal evaluation in this study [17]. It is a widely used public benchmark for heartbeat classification and contains 48 half-hour ambulatory ECG recordings with expert beat annotations. Following the AAMI recommendation, the original beat annotations were grouped into four heartbeat classes: N, S, V, and F. The corresponding mapping between the original annotation symbols and the four heartbeat classes used in this study is presented in table 1.

In addition to the primary MIT-BIH-based experiments, an independent external validation experiment was conducted on the INCART Arrhythmia Database [18, 19] to further assess cross-dataset generalization. The INCART dataset was not used for model training, validation-based model selection, or ablation analysis; it was used only for independent external testing. For the MIT-BIH dataset, the MLI lead was used for subsequent preprocessing and beat-level classification, because it provides relatively clear waveform morphology for heartbeat analysis.

Following the AAMI standard for heartbeat classification, the original beat annotations were mapped into four heartbeat classes used in this study. Specifically, class N includes normal and bundle branch block related beats (N, L, R, e, j), class S includes supraventricular ectopic beats (A, a, J, S), class V includes ventricular ectopic beats (V, E), and class F denotes fusion beats (F).

The MIT-BIH Arrhythmia Database used in this study exhibits substantial class imbalance, in which the minority heartbeat classes occur much less frequently than the dominant normal class. To avoid data leakage, the dataset was first partitioned using a patient-level split, so that all heartbeat samples from the same subject were assigned to only one subset. After dataset splitting, SMOTE was applied only to the training set to alleviate class imbalance, whereas the validation and test sets retained their original

Table 1. Mapping of original beat annotations to the four AAMI heartbeat classes used in this study.

Type of ECG signal	Specific classification
Normal ECG signal (N)	Normal heartbeat (N) Left bundle branch block (L) Right bundle branch block (R) Atrial escape (e) Nodal/junctional escape beat (j)
Supraventricular ectopia (S)	Atrial premature Beat (A) Abnormal atrial premature Beat (a) Nodal/junctional premature beat (J) Supraventricular premature beat (S)
Ventricular ectopic (V)	Ventricular premature beat (V) Ventricular escape (E)
Fused ECG signal (F)	Ventricular fusion beats (F)

class distributions. The original validation and test subsets were kept unchanged during evaluation, and the post-SMOTE training samples were used only for model training. Therefore, post-SMOTE sample counts were not mixed with the original validation or test distributions. Instead, the data handling strategy is stated explicitly in the text, and all internal evaluation results on the MIT-BIH dataset are based on unchanged validation and test subsets.

2.2. Related work on ECG beat classification

Automated ECG beat classification has been widely studied for computer-aided arrhythmia diagnosis. Early studies mainly relied on handcrafted features, such as morphological descriptors, RR-interval features, wavelet-domain coefficients, entropy indices, and statistical features, combined with conventional classifiers including SVMs, random forests, k -nearest neighbors, and shallow neural networks [20]. Although these methods are interpretable and computationally efficient, their performance strongly depends on manual feature design and may be limited by inter-patient morphological variability, noise interference, and severe class imbalance [20].

With the development of DL, CNN-based models have become mainstream for ECG classification because they can automatically learn hierarchical waveform representations from raw or preprocessed ECG signals [20]. However, conventional CNNs usually rely on fixed receptive fields, which may be insufficient to characterize arrhythmia-related patterns distributed across different temporal scales. Therefore, multi-scale convolutional and hierarchical feature extraction structures have been introduced to capture both local waveform transitions and broader morphological configurations [21–23]. These methods improve the representation of heterogeneous ECG morphologies, especially for supraventricular ectopic beats, ventricular ectopic beats, and fusion beats.

In addition to morphological feature extraction, temporal dependency modeling is also important for ECG beat classification. Hybrid CNN-RNN frameworks combine CNN-based waveform representation with LSTM, BiLSTM, or related recurrent structures to model sequential dependencies in extracted feature sequences [24, 25]. Meanwhile, attention mechanisms, Transformer-based models, and emerging architectures such as KAN and xLSTM have been introduced to enhance discriminative feature selection and long-range dependency modeling [26–28]. Nevertheless, these models often increase architectural complexity and may require more careful evaluation under patient-wise and cross-dataset settings.

Despite substantial progress, several challenges remain. First, public ECG datasets are highly imbalanced, and minority classes such as supraventricular ectopic and fusion beats are difficult to identify reliably [29]. Although SMOTE, focal loss, data augmentation, and hybrid sampling have been used to alleviate class imbalance [29], oversampling must be applied only to the training set after patient-wise splitting to avoid information leakage. Second, inter-patient variability and cross-dataset differences in lead configuration, sampling frequency, annotation criteria, and population characteristics still limit model generalization. Third, many high-performance models have relatively large parameter sizes, which motivates lightweight ECG classification designs for wearable and embedded monitoring scenarios [30].

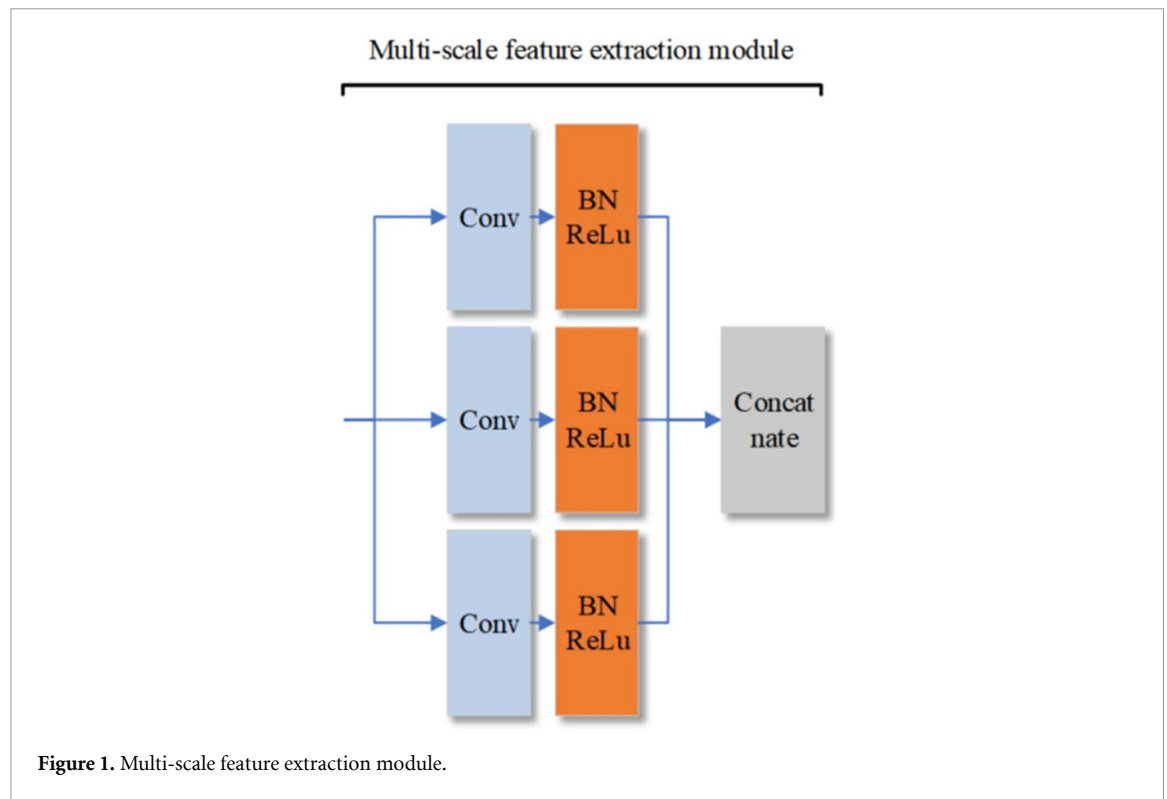


Figure 1. Multi-scale feature extraction module.

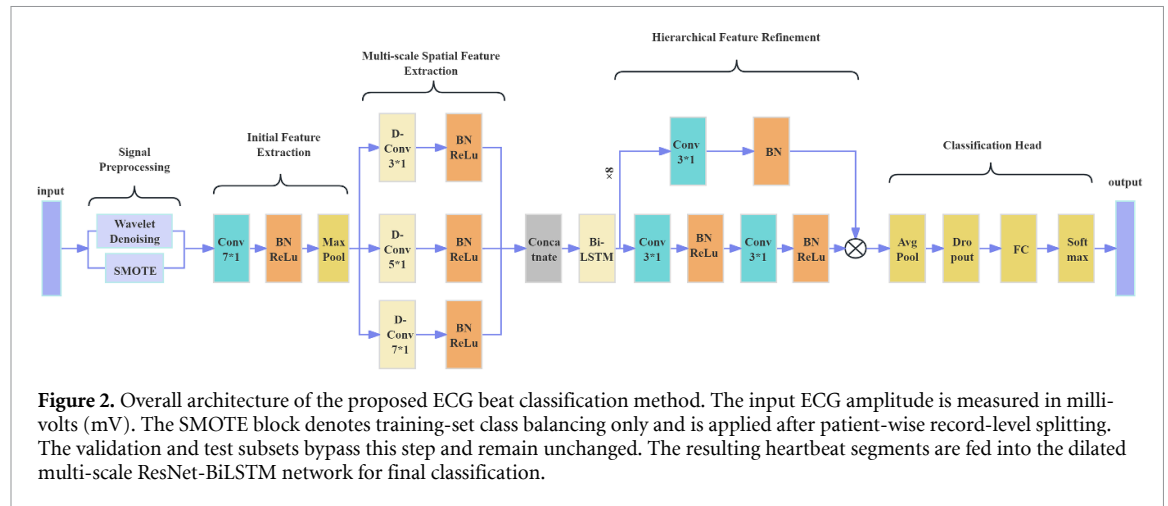
Motivated by these limitations, this study proposes a novel ECG beat classification framework. Different from many conventional CNN-based or unidirectional temporal modeling methods, the proposed method integrates a dilated multi-scale residual structure with BiLSTM-based temporal feature modeling to jointly capture morphological details and temporal dependencies.

3. ECG beat classification framework based on dilated multi-scale ResNet and BiLSTM

The information contained in ECG signals is highly complex and mainly involves waveform morphology and temporal dependency. To better characterize ECG beat morphology at different temporal scales, this study designs a multi-scale feature extraction module inspired by the Inception concept. Three convolution kernels of different sizes (3×1 , 5×1 , and 7×1) are introduced in parallel to capture waveform patterns at different scales. As shown in figure 1, the proposed module consists of three branches, each including convolution, normalization, and nonlinear activation. The outputs of the three branches are concatenated along the channel dimension to form a unified representation, thereby improving feature extraction efficiency while preserving multi-scale morphological information.

In the proposed framework, one-dimensional convolution operates by sliding learnable kernels along the temporal axis of the ECG signal. During this process, each kernel produces a strong response when the local waveform segment matches its learned pattern. Therefore, the convolutional layer does not explicitly identify the P, Q, R, S, and T points in a rule-based manner; instead, it automatically learns local morphological patterns associated with the P-QRS-T complex, such as slope changes, sharp peaks, wave width variations, and short-range waveform configurations. Small kernels are more sensitive to fine local details, such as the sharp transition around the QRS complex, whereas larger kernels can capture relatively broader waveform structures, such as the shape evolution from the P wave to the QRS complex or from the QRS complex to the T wave. In this way, convolutional feature extraction enables the model to represent local ECG morphology at multiple temporal scales.

Although the multi-scale residual backbone is effective for capturing morphological information at different temporal scales, it does not explicitly model sequential dependency within the ECG heartbeat segment. In addition, the introduction of multi-scale branches increases model complexity and parameter quantity. To address this limitation, a BiLSTM layer is incorporated into the proposed framework to capture temporal dynamics from the aggregated multi-scale representation. By combining multi-scale convolutional feature extraction with bidirectional temporal modeling, the proposed model is expected to improve ECG beat classification while maintaining favorable model efficiency.



The system flow of the method is shown in figure 2. At the front end of the network is the convolutional layer, followed immediately by the multi-scale layer. The multi-scale layer consists of three parallel dilated convolutions with kernel sizes of 3×1 , 5×1 , and 7×1 , respectively. They are responsible for extracting multi-scale morphological features from the ECG waveform. Additionally, batch normalization layers are integrated after the dilated convolution to expedite the training speed of the network, while ReLU activation functions are included to prevent gradient vanishing. After the multi-scale layer, the aggregation layer combines the extracted multi-scale waveform features and feeds them into the BiLSTM for temporal dependency modeling. This is then followed by the residual layer. Finally, after average pooling and flattening, the features are fed into the fully connected layer for classification. As shown in figure 2, the proposed architecture integrates ECG preprocessing, dilated multi-scale convolutional feature extraction, BiLSTM-based temporal modeling, hierarchical feature refinement, and final classification. In this study, SMOTE is not treated as a general preprocessing operation applied to all data subsets. Instead, it is performed only after patient-wise record-level splitting and only on the training heartbeat samples. The validation and test subsets remain unchanged during evaluation.

It should be noted that the SMOTE block shown in figure 2 is a schematic representation of the training-set class-balancing strategy rather than a universal preprocessing step. In the actual implementation, SMOTE is applied only to the training heartbeat samples after patient-wise record-level splitting, whereas the validation and test subsets are not oversampled, resampled, or modified.

3.1. Overall framework

The proposed ECG beat classification framework is designed to jointly exploit multi-scale morphological features and bidirectional temporal dependencies. As shown in figure 2, the model adopts ResNet18 as the backbone due to its stable optimization enabled by residual connections and its capability to extract hierarchical waveform features from local to global scales. To effectively capture the temporal dynamics within the heartbeat sequence, a BiLSTM network is integrated into the architecture. Unlike unidirectional recurrent structures, the BiLSTM layer processes the feature sequence in both forward and backward directions, allowing the model to leverage contextual information from preceding and subsequent time steps within the extracted feature sequence for more robust classification.

The overall architecture of the proposed method is illustrated in figure 2. Given an input ECG beat segment x , the model first performs initial convolutional feature extraction to obtain a low-level representation h_0 . Then, a multi-scale dilated residual module is used to extract morphological features at different temporal scales through three parallel branches with kernel sizes of 3×1 , 5×1 , and 7×1 . The resulting multi-scale features are concatenated and further fed into a BiLSTM layer to capture bidirectional temporal dependencies within the extracted heartbeat feature sequence. Finally, the output representation is passed through global average pooling and a fully connected layer with Softmax activation to generate the final heartbeat class prediction.

According to this processing pipeline, the proposed framework consists of three main stages: (1) initial feature extraction from the input ECG beat, (2) multi-scale morphological representation with dilated convolution, and (3) bidirectional temporal dependency modeling followed by classification. This design enables the model to jointly exploit local waveform morphology, multi-scale structural information, and temporal rhythm context in a unified framework.

Through this design, the proposed framework jointly exploits multi-scale morphology representation, bidirectional temporal context, and parameter-efficient receptive field expansion for ECG beat classification. The learned representations can be interpreted at three levels. First, the early convolutional layers mainly encode local morphological features, such as slope changes, sharp peaks, and wave width variations, related to the P-QRS-T complex. Second, the multi-scale branches provide structural features at different temporal scales, capturing both fine details of local transitions and broader waveform configurations. Finally, the BiLSTM layer models temporal dependencies within the extracted feature sequence, enabling the model to interpret each heartbeat segment using both forward and backward temporal feature information.

3.2. Data preprocessing

For each recording, the ECG signal from the selected MLI lead was used for subsequent preprocessing and beat-level classification. The preprocessing pipeline consisted of baseline-drift suppression, wavelet-based denoising, heartbeat segmentation, patient-level dataset splitting, and training-set class balancing. Specifically, median filtering was first used to estimate and suppress baseline drift in the raw ECG signal. Subsequently, discrete wavelet denoising based on the db5 wavelet was applied to the baseline-corrected signal to further reduce noise.

An 8-level wavelet decomposition was adopted in this study because the MIT-BIH recordings were sampled at 360 Hz, and this decomposition depth provides a suitable frequency resolution for separating very-low-frequency baseline drift and high-frequency noise from diagnostically relevant ECG components. In particular, the deeper approximation component mainly captures the low-frequency baseline fluctuation, whereas the detail components preserve the morphological information of the P wave, QRS complex, and T wave. Empirically, this setting provided a favorable trade-off between denoising effectiveness and waveform preservation in our preliminary preprocessing trials. After denoising, heartbeat samples were segmented around annotated beat locations. A representative example of the denoising result is shown in figure 3. The class-balancing procedure applied to the training set is illustrated in figure 4. The complete workflow of the proposed ECG preprocessing and classification pipeline is presented in algorithm 1.

The preprocessing procedure used in this study can be summarized as follows. Let $s(t)$ denote the raw ECG recording from the selected lead. First, baseline drift is suppressed by median filtering, and wavelet denoising is then applied to obtain a denoised signal $sd(t)$. Second, heartbeat samples are segmented around the annotated beat locations r_i , and the i th heartbeat segment is defined as:

$$x_i = sd[r_i - l : r_i + r], i = 1, 2, \dots, N \quad (1)$$

where $l = 99$ and $r = 160$ denote the numbers of samples taken before and after the annotated beat location, respectively. After heartbeat segmentation and AAMI class mapping, the MIT-BIH Arrhythmia Database was partitioned at the record level to prevent patient-level information leakage. Each ECG record was regarded as an independent patient record, and all heartbeat samples from the same record were assigned to only one subset. Specifically, the 48 MIT-BIH records were divided into 28 training records, 9 validation records, and 11 test records, corresponding approximately to a 60%/20%/20% split. No record overlap existed among the training, validation, and test subsets. The record IDs used in each subset are listed in table 2. The held-out test set consisted of records 101, 105, 106, 107, 108, 115, 116, 118, 214, 219, and 222. No beat-level or segment-level selection was applied to the test set. All annotated beats in the selected test records that could be mapped to the four AAMI heartbeat classes, namely N, S, V, and F, were used for final evaluation. The test set was not manually filtered, artificially screened, resampled, or modified. After patient-wise record-level partitioning, SMOTE was applied only to the training set to alleviate class imbalance. The validation and test sets retained their original class distributions. This procedure ensures that synthetic samples generated by SMOTE were not introduced into the validation or test subsets, thereby avoiding information leakage caused by oversampling.

To further clarify the composition of the held-out test set, the original beat counts of each test record after AAMI class mapping are reported in table 3. These counts were calculated before SMOTE. Since SMOTE was applied only to the training set, the beat counts of the test records remained unchanged during all experiments.

The beat counts in table 3 were recalculated from the original MIT-BIH annotation files after AAMI class mapping and before SMOTE. Since SMOTE was applied only to the training set, the held-out MIT-BIH test records remained unchanged during evaluation. Therefore, the class-wise Support values used in the performance evaluation are consistent with the total N, S, V, and F counts reported in table 3.

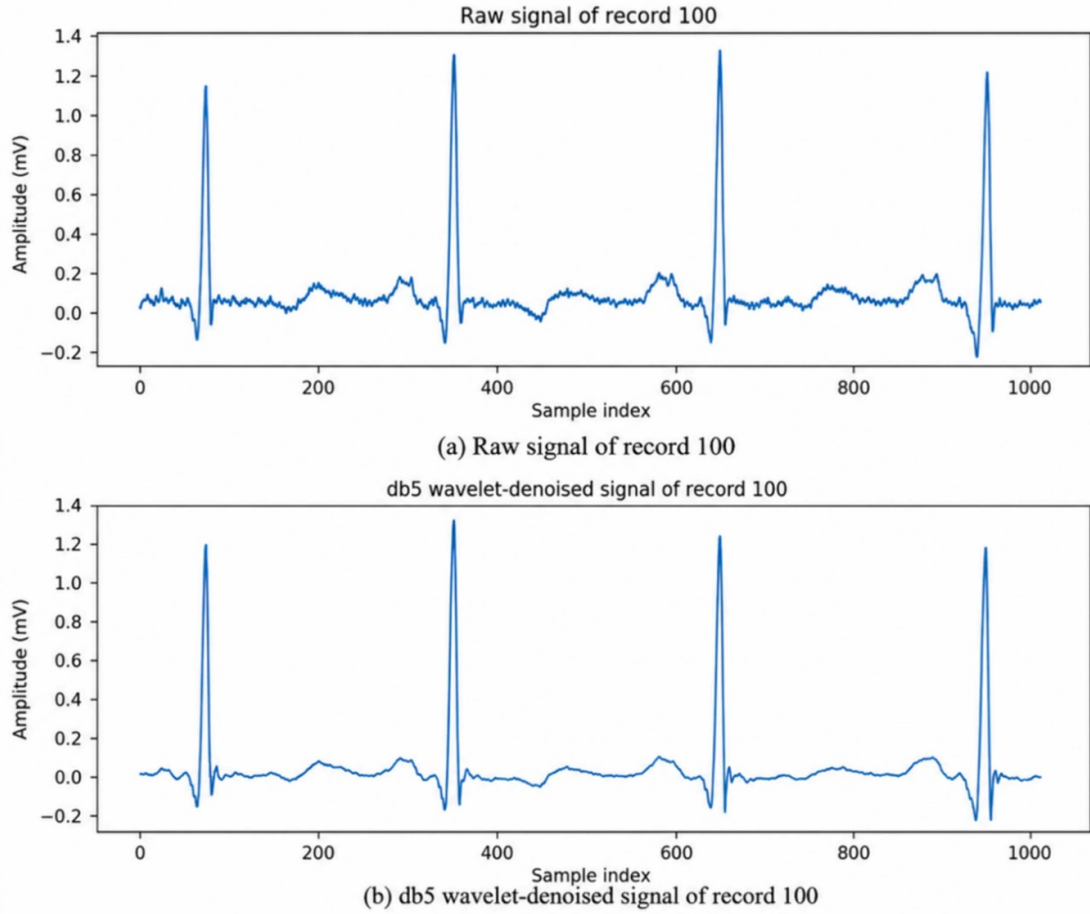


Figure 3. Denoising result of record 100 using the db5 wavelet.

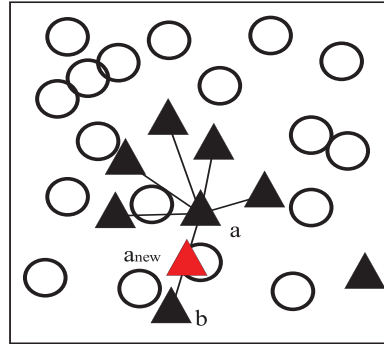


Figure 4. Schematic diagram of the SMOTE-based class balancing strategy applied only to the training set.

3.3. Integration of dilated convolution

In the conventional multi-scale convolutional design, enlarging the receptive field usually relies on increasing the kernel size or stacking more convolutional layers, which leads to additional parameters and computational burden. To address this issue, the standard convolutions in the multi-scale branches are replaced with one-dimensional dilated convolutions along the temporal axis.

For a 1-D dilated convolution, the output at position (t) can be expressed as:

$$y(t) = \sum_{i=0}^{K-1} w(i) x(t - d \cdot i) \quad (2)$$

where K is the kernel size, d is the dilation rate, and $w(i)$ denotes the convolution kernel weights. Compared to standard convolution, dilated convolution enlarges the effective receptive field without

Algorithm 1. Proposed ECG beat classification algorithm.

Input: Raw ECG records R , expert annotations A , patient-wise record partitions S_{train} , S_{val} , and S_{test} , maximum epoch E_{max}
Output: Predicted labels and evaluation metrics on the held-out test set

- 1: Initialize the model f_{θ} and the Adam optimizer.
- 2: for each subset $S \in \{S_{\text{train}}, S_{\text{val}}, S_{\text{test}}\}$ do
- 3: Initialize the heartbeat sample set $D_S \leftarrow \emptyset$.
- 4: for each ECG record $r \in S$ do
- 5: Select the target lead signal x_r .
- 6: Remove baseline drift from x_r using median filtering.
- 7: Apply db5 wavelet denoising to obtain the denoised signal x'_r .
- 8: for each annotated beat location a_i in record r do
- 9: Map the original annotation to the corresponding AAMI class y_i .
- 10: if $y_i \in \{N, S, V, F\}$ and the heartbeat segment is valid then
- 11: Segment the heartbeat sample X_i around a_i using 99 pre-beat and 160 post-beat samples.
- 12: Add (X_i, y_i) to D_S .
- 13: end if
- 14: end for
- 15: end for
- 16: end for
- 17: Apply SMOTE only to D_{train} to obtain the balanced training set $D_{\text{train}}^{\text{bal}}$.
- 18: Keep D_{val} and D_{test} unchanged.
- 19: for epoch = 1 to E_{max} do
- 20: for each mini-batch (X_b, y_b) sampled from $D_{\text{train}}^{\text{bal}}$ do
- 21: Compute the predicted probability $p_b = f_{\theta}(X_b)$.
- 22: Calculate the cross-entropy loss L between p_b and y_b .
- 23: Update θ using the Adam optimizer.
- 24: end for
- 25: Evaluate the current model on D_{val} .
- 26: if the validation loss decreases then
- 27: Save the current model parameters θ^* .
- 28: end if
- 29: end for
- 30: Load the best model parameters θ^* .
- 31: Evaluate f_{θ^*} on the unchanged held-out test set D_{test} .
- 32: Return the predicted labels, confusion matrix, accuracy, precision, recall, and F1 score.

Table 2. Patient-wise record partition of the MIT-BIH Arrhythmia Database.

Subset	No. of records	Record IDs
Training records	28	103, 104, 109, 111, 112, 113, 119, 122, 123, 124, 200, 201, 202, 203, 207, 208, 209, 210, 212, 213, 215, 217, 221, 223, 230, 231, 233, 234
Validation records	9	100, 102, 114, 117, 121, 205, 220, 228, 232
Test	11	101, 105, 106, 107, 108, 115, 116, 118, 214, 219, 222

increasing the number of learnable weights in each filter, and is applied to the MIT-BIH dataset in our experiments.

In the proposed framework, three parallel dilated convolution branches with kernel sizes 3×1 , 5×1 , and 7×1 are used to capture ECG waveform patterns at different temporal scales. The output of each branch can be written as:

$$h_k = \text{ReLU}(\text{BN}(\text{DConv}_{k,d_k}(h_0))), \quad k \in \{3, 5, 7\} \quad (3)$$

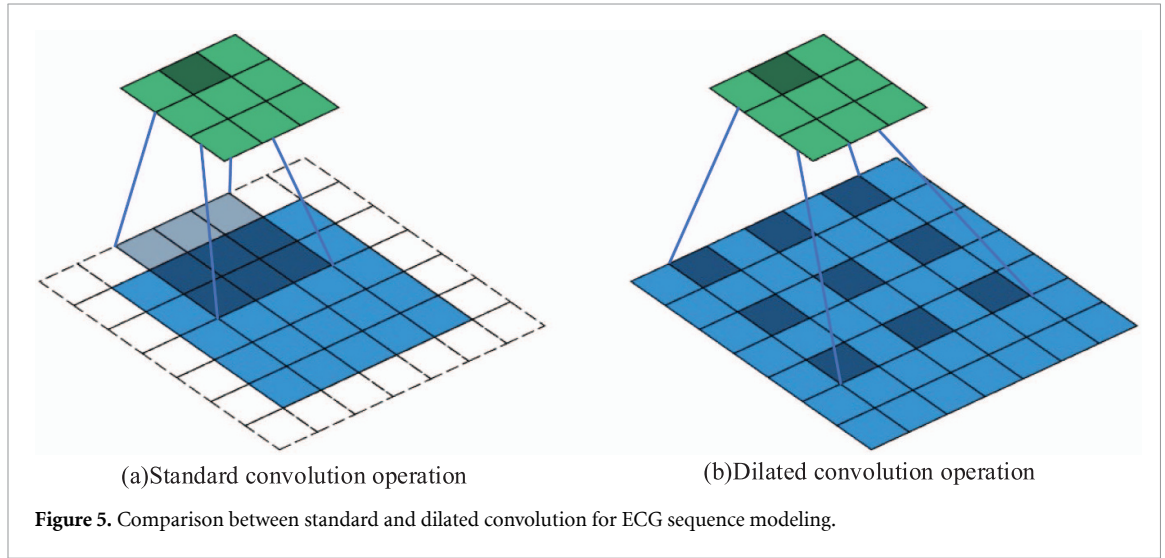
where h_0 denotes the initial feature representation extracted by the backbone, and Conv_{k,d_k} denotes a dilated convolution with kernel size k and dilation rate d_k . The outputs of the three branches are then aggregated as:

$$h_{\text{ms}} = \text{Concat}(h_3, h_5, h_7). \quad (4)$$

Here, $h_0 \in \mathbb{R}^{C \times T}$ denotes the feature map generated by the initial convolutional backbone, where C is the number of channels and T is the temporal length. The outputs h_3 , h_5 , and h_7 correspond to the

Table 3. Beat counts of each record in the held-out MIT-BIH test set before SMOTE.

Record ID	N	S	V	F	Total
101	1859	3	0	0	1862
105	2526	0	41	0	2567
106	1507	0	520	0	2027
107	0	0	59	0	59
108	1739	4	17	2	1762
115	1952	0	0	0	1952
116	2301	1	109	0	2411
118	2165	96	16	0	2277
214	2001	0	256	1	2258
219	2082	7	64	1	2154
222	2272	209	0	0	2481
Total	20404	320	1082	4	21810



three parallel dilated branches with different kernel sizes. After concatenation, the aggregated representation h_{ms} contains complementary morphological information extracted at different temporal scales and serves as the input to the subsequent BiLSTM module.

As shown in figure 5, compared with standard convolution, dilated convolution enables the network to capture both local waveform morphology and broader rhythm context more effectively. This design not only improves the representation of heterogeneous ECG beats, but also helps control parameter growth compared with the conventional BiLSTM-ResNet design. Therefore, compared with BiLSTM-ResNet, the proposed CBiLSTM-ResNet provides a better balance between classification performance and model efficiency.

3.4. BiLSTM-based temporal modeling

After multi-scale feature extraction and feature aggregation, the concatenated representation is used as the input to the BiLSTM layer. The concatenated multi-scale representation is defined as equation (5):

$$h_{ms} = \text{Concat}(h_3, h_5, h_7) \quad (5)$$

where h_3 , h_5 , and h_7 represent the outputs of the three convolutional branches with different kernel sizes.

To model temporal dependency in ECG beat sequences, the concatenated representation in equation (5) is then fed into a BiLSTM network. The forward hidden state is computed as:

$$h_t^f = \text{LSTM}_f(h_{ms,t}, h_{t-1}^f) \quad (6)$$

and the backward hidden state is computed as:

$$h_t^b = \text{LSTM}_b(h_{ms,t}, h_{t+1}^b). \quad (7)$$

The final hidden representation at time step t is obtained by concatenating the forward and backward hidden states:

$$h_t = \text{Concat}(h_t^f, h_t^b) \quad (8)$$

where h_t^f and h_t^b denote the forward and backward hidden states at time step t , respectively. Compared with a unidirectional LSTM, the BiLSTM layer can simultaneously exploit contextual information from both preceding and subsequent time steps in the extracted feature sequence. This is particularly useful for ECG beat classification because abnormal heartbeat segments may contain morphological variations distributed across different temporal positions, such as changes in waveform slope, QRS width, and local waveform configuration. Therefore, bidirectional temporal modeling is more suitable for distinguishing morphologically similar heartbeat classes. After BiLSTM-based temporal modeling, the context-enhanced feature sequence is further passed through a residual convolutional refinement block, as shown in figure 2. This block further integrates local and sequential representations before global average pooling and final classification.

The final sequence representation is then passed to the classification head:

$$\hat{y} = \text{Softmax}(W_{fc} \cdot \text{GAP}(H) + b_{fc}) \quad (9)$$

where $\text{GAP}(\cdot)$ denotes global average pooling, H denotes the final sequence representation, and W_{fc} and b_{fc} are the parameters of the fully connected layer. In this process, the BiLSTM receives the aggregated multi-scale feature sequence as input and produces a context-enhanced representation by combining the forward and backward hidden states at each time step. The final representation H is then compressed by global average pooling and mapped to the output space through the fully connected layer. Therefore, the classification head operates on a temporally enriched feature representation rather than on independent local features only. According to equation (9), the final heartbeat class prediction is obtained by applying the fully connected layer and the Softmax function to the final sequence representation. In this way, the BiLSTM layer enhances beat classification by explicitly modeling bidirectional temporal dependency after multi-scale morphological feature extraction.

The cross-entropy loss function is adopted for the multi-class ECG beat classification task. The loss function measures the discrepancy between the predicted probability distribution and the true one-hot label distribution, and is defined as:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^4 y_{ic} \log(\hat{y}_{ic}) \quad (10)$$

where N denotes the total number of heartbeat samples in a mini-batch, y_{ic} is the true one-hot label of category c for the i th sample, and $\hat{y}_{ic}$ is the corresponding predicted probability output by the Softmax layer in equation (9). The cross-entropy loss is suitable for this four-class classification problem and guides the model to learn discriminative features for different arrhythmia types during training.

4. Experiment and analysis

4.1. Experimental settings

To improve reproducibility, the main experimental settings used in this study are summarized in table 4.

For fair comparison, all compared models were trained using the same data split, preprocessing procedure, and optimization settings unless otherwise specified.

The evaluation metrics used in this study include the confusion matrix, recall (Rec), precision (Pre), $F1$ score, and overall accuracy (Acc). These metrics were used to assess the classification performance of the model across the four heartbeat classes, namely N, S, V, and F.

4.2. Selection and comparative analysis of wavelet basis function

To validate the rationality of selecting the db5 wavelet as the basis function, we conducted a comparative evaluation among three widely adopted wavelets, db2, db5 and db8, using three representative ECG records, 100, 119 and 208, from the MIT-BIH arrhythmia database. All three records (100, 119, and 208) were assigned to the development set, which consisted of the training and validation partitions, and none of them belonged to the held-out test set. The quantitative assessment metrics consisted of signal-to-noise ratio (SNR), root mean square error (RMSE), and R-wave amplitude retention rate.

Table 4. Experimental settings used in this study.

Category	Item	Setting
Hardware/software	Operating system	Windows 10
	CPU	Intel Core i7-12700
	GPU	NVIDIA RTX 3080
	CUDA version	11.8
	Framework	PyTorch 1.10
	Python version	3.9
Data splitting	Dataset	MIT-BIH Arrhythmia Database
	Split strategy	Patient-wise record-level split
	Train/validation/test ratio	28/9/11 records, approximately 60%/20%/20%
	Data leakage prevention	No record overlap among training, validation, and test subsets
	Class balancing	SMOTE applied only to the training set
	Random seed	42
Preprocessing	Denoising method	Median filtering, wavelet
	Heartbeat segmentation window	99 samples before, 160 after
Training	Batch size	64
	Number of epochs	50
	Optimizer	Adam, weight decay = 1×10^{-5}
	Initial learning rate	0.001
	Learning rate schedule	Step decay every 20 epochs
	Loss function	Cross-entropy Loss
	Dropout	0.5
Model selection	Best model criterion	Lowest validation loss
Evaluation	Evaluation metrics	Accuracy, recall, precision, <i>F1</i>

Table 5. Denoising performance comparison using different wavelet basis functions.

Wavelet	SNR (dB)	RMSE	R-wave retention
db2	12.39	0.7500	0.9120
db5	14.73	0.6206	0.9619
db8	15.20	0.5795	0.9368

These metrics were used jointly to evaluate the trade-off between noise suppression and ECG morphological preservation. The average experimental results are summarized in table 5.

The experimental results demonstrate that the db5 wavelet provides a favorable balance between noise suppression and ECG morphological preservation. Although db8 achieved the highest SNR and the lowest RMSE, its R-wave retention rate was lower than that of db5, indicating a stronger smoothing effect that may weaken diagnostically important waveform details. In contrast, db5 achieved the highest R-wave retention rate while maintaining competitive SNR and RMSE values. Therefore, db5 was selected as the wavelet basis function because it provided a better trade-off between denoising performance and ECG morphology preservation.

4.3. Ablation studies

To verify the contribution of each component in the proposed framework, a series of ablation experiments were conducted on the MIT-BIH dataset under the same preprocessing, training, and model-selection protocol. The compared models include the baseline ResNet, M-ResNet with the multi-scale module, BiLSTM-ResNet with both the multi-scale module and BiLSTM layer, and CBiLSTM-ResNet with additional dilated convolution. The validation set was used only for model selection, and the held-out test set was used only for final evaluation. To address the reviewer's concern regarding numerical consistency, the class-wise statistics were recalculated using the same unchanged held-out MIT-BIH test set. The Support value of each heartbeat class was kept identical across all compared models and was made consistent with the test-set beat counts reported in table 3.

Table 6. Class-wise performance comparison of different models on the same unchanged MIT-BIH test set.

Model	Class	Support	TP	FN	FP	Precision	Recall	F1score
ResNet	N	20404	20170	234	177	99.13%	98.85%	98.99%
	S	320	220	100	110	66.67%	68.75%	67.69%
	V	1082	957	125	116	89.19%	88.45%	88.82%
	F	4	1	3	59	1.67%	25.00%	3.12%
M-ResNet	N	20404	20234	170	101	99.50%	99.17%	99.33%
	S	320	265	55	75	77.94%	82.81%	80.30%
	V	1082	972	110	95	91.10%	89.83%	90.46%
	F	4	3	1	65	4.41%	75.00%	8.33%
BiLSTM-ResNet	N	20404	20274	130	61	99.70%	99.36%	99.53%
	S	320	290	30	60	82.86%	90.62%	86.57%
	V	1082	1025	57	70	93.61%	94.73%	94.17%
	F	4	3	1	27	10.00%	75.00%	17.65%
CBiLSTM-ResNet	N	20404	20314	90	26	99.87%	99.56%	99.72%
	S	320	305	15	40	88.41%	95.31%	91.73%
	V	1082	1055	27	45	95.91%	97.50%	96.70%
	F	4	3	1	22	12.00%	75.00%	20.69%

4.3.1. Component-wise ablation results

For clarity, the baseline model is denoted as ResNet, the model with the multi-scale module as M-ResNet, the model combining the multi-scale module and BiLSTM as BiLSTM-ResNet, and the final model further incorporating dilated convolution as CBiLSTM-ResNet. This baseline was chosen because it provides a stable and structurally clear backbone for evaluating the contribution of each subsequent module.

To verify the contributions of the multi-scale module and the BiLSTM layer to ECG beat classification, four models were compared in this subsection: ResNet, M-ResNet, BiLSTM-ResNet, and CBiLSTM-ResNet. Here, M-ResNet denotes the model obtained by introducing the multi-scale module into the baseline ResNet, and BiLSTM-ResNet denotes the model obtained by further incorporating a BiLSTM layer into M-ResNet. The effect of introducing dilated convolution is discussed separately through the CBiLSTM-ResNet model.

All models in table 6 were evaluated on the same unchanged held-out MIT-BIH test set described in table 3. Therefore, for each heartbeat class, the Support value, defined as $TP + FN$, remains identical across all compared models and is consistent with the class-wise totals reported in table 3. To improve numerical transparency, table 6 reports Support, TP, FN, FP, Precision, Recall, and F1 score for each heartbeat class.

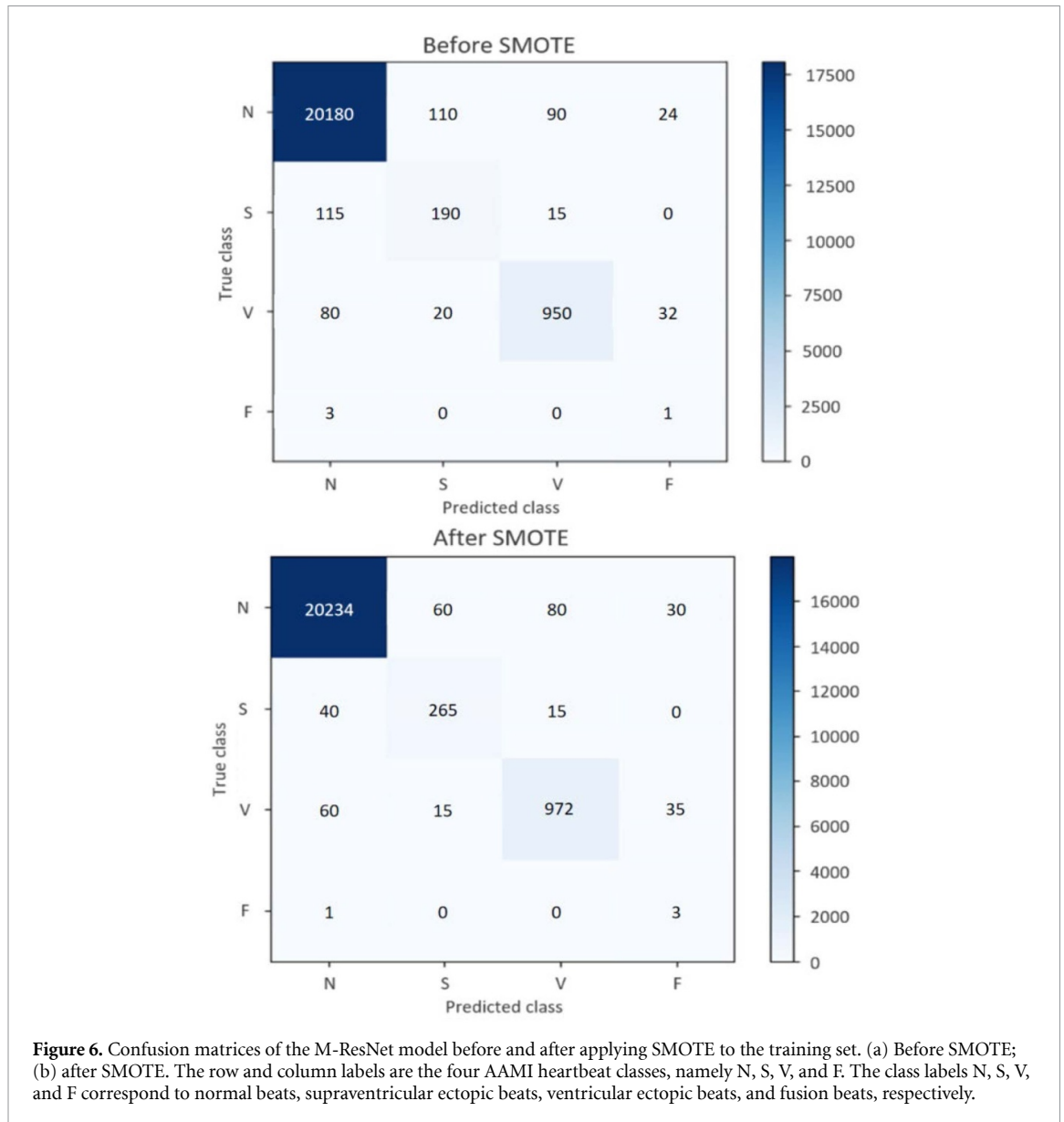
Based on the same unchanged MIT-BIH test set, the overall accuracies of ResNet, M-ResNet, BiLSTM-ResNet, and CBiLSTM-ResNet were 97.88%, 98.46%, 99.00%, and 99.39%, respectively. It should be noted that the F class contains only four samples in the held-out MIT-BIH test set. Therefore, the class-wise Precision and F1 score of F beats should be interpreted cautiously. Overall, the performance improvement is mainly reflected in the N, S, and V classes, while the F-class results are reported only as supplementary class-wise evidence due to the extremely limited number of F beats.

4.3.2. Effect of class balancing strategy

Because the MIT-BIH Arrhythmia Database exhibits severe class imbalance, especially for the S and F classes, the effect of training-set class balancing was further examined. In this experiment, SMOTE was applied only to the training set after patient-wise splitting, while the validation and test sets were kept unchanged. This design prevents synthetic samples from contaminating the evaluation subsets.

The confusion matrices used to illustrate the effect of applying SMOTE to the training set are shown in figure 6. To avoid ambiguity, the axes of the confusion matrices are labeled using the four AAMI heartbeat classes, namely N, S, V, and F. The upper matrix represents the result before SMOTE, whereas the lower matrix represents the result after applying SMOTE only to the training set.

Figure 6 is used to illustrate the influence of applying SMOTE to the training set on class-balancing behavior in the ablation experiment. The row sums of the confusion matrices are consistent with the class-wise test-set composition reported in table 3. As shown in figure 6, applying SMOTE only to the training set improves the recognition of minority classes, especially S beats, indicating that training-set class balancing helps alleviate the bias toward the dominant N class. For the F class, although the



number of correctly classified beats increases after SMOTE, the result should be interpreted cautiously because only four F beats are included in the held-out test set. Therefore, SMOTE is used here as a standard preprocessing strategy to improve minority-class recognition, rather than as a methodological innovation or a direct guarantee of improved overall accuracy.

4.3.3. Effect of dilated convolution

The integration of dilated convolution is intended to maintain classification accuracy while controlling parameter growth, thereby improving model efficiency. The integration of dilated convolution with the BiLSTM-ResNet network is named CBiLSTM-ResNet. It is then tested on the same test set. The number of parameters for each model is calculated, and the results are shown in table 7. The accuracy values of ResNet, BiLSTM-ResNet, and CBiLSTM-ResNet in table 7 are consistent with the overall accuracies calculated from the class-wise results reported in table 6.

Table 7 compares the model complexity and classification accuracy of the proposed ablation models and representative methods reported in the literature. For the proposed models, all accuracy values were obtained from the same unchanged held-out MIT-BIH test set used in the ablation experiments. Compared with the baseline ResNet, BiLSTM-ResNet improves the classification accuracy from 97.88% to 99.00%, indicating that temporal feature modeling contributes to ECG beat classification performance. However, this improvement is accompanied by an increase in the number of parameters from 25.8 M to 35.3 M.

Table 7. Accuracy and parameter comparison of proposed variants and representative literature methods.

Model	Source	Signal dimension	Parameters (M)	Accuracy
ResNet	Proposed	1D	25.8	97.88%
BiLSTM-ResNet	Proposed	1D	35.3	99.00%
CBiLSTM-ResNet	Proposed	1D	27.8	99.39%
RNN	Übeyli [31]	1D	N/A	98.94%
MSCNN-Seq2Seq	Khatar <i>et al</i> [32]	1D	N/A	98.94%
LNN	Liu <i>et al</i> [33]	1D	N/A	99.24%

Table 8. Comparison with recent representative studies on ECG classification.

References	Year	Method	Dataset	Main reported metric	Reported result	Parameters
Wang <i>et al</i> [14]	2021	CNN + NCBAM	MIT-BIH Arrhythmia Database	F1 score	96.64%	N/A
Khatar <i>et al</i> [32]	2024	MSCNN-Seq2Seq	Reported in the original paper	Accuracy	98.94%	N/A
Proposed method		Dilated multi-scale ResNet + BiLSTM	MIT-BIH Arrhythmia Database	Accuracy	99.39%	27.8 M

It should be noted that dilated convolution itself does not reduce the number of learnable kernel weights when the kernel size and channel configuration remain unchanged. Its main advantage is to enlarge the effective receptive field without requiring larger convolution kernels or additional stacked convolutional layers. In the proposed CBiLSTM-ResNet, the dilated convolution design enables a more compact feature extraction structure than the BiLSTM-ResNet variant while preserving a broad temporal receptive field. As a result, the total number of parameters is reduced from 35.3 M to 27.8 M compared with BiLSTM-ResNet, corresponding to a reduction of 21.25%, while the classification accuracy further increases to 99.39%. Compared with the baseline ResNet, CBiLSTM-ResNet improves the accuracy by 1.51 percentage points, with only a moderate increase in parameter size from 25.8 M to 27.8 M. These results suggest that the proposed architecture provides a favorable balance between receptive-field expansion, classification performance, and model complexity.

It should be noted that the results of RNN, MSCNN-Seq2Seq, and LNN are directly taken from their original publications. Because different studies may adopt different preprocessing strategies, dataset partitions, class mappings, and evaluation protocols, this comparison is intended to provide a literature-level reference rather than a strictly controlled head-to-head benchmark. Under this setting, the proposed CBiLSTM-ResNet achieves competitive performance and provides a favorable balance between classification accuracy and model complexity.

4.4. Comparison with recent representative studies

Although the above ablation experiments verify the effectiveness of the proposed components within our framework, they are limited to internal model variants. To further evaluate the competitiveness of the proposed method, we compared it with representative ECG beat classification studies published in recent years. Since different studies may adopt different preprocessing strategies, data partition protocols, class mappings, and evaluation metrics, this comparison is intended to provide a literature-level reference rather than a strictly controlled head-to-head benchmark. Whenever possible, results reported on the MIT-BIH Arrhythmia Database are prioritized for comparison.

As shown in table 8, the proposed method achieves an accuracy of 99.39% on the MIT-BIH Arrhythmia Database, which is competitive with representative ECG classification methods reported in recent studies. It should be noted that different studies may use different preprocessing procedures, data partition strategies, class definitions, and evaluation protocols. Therefore, the comparison in table 8 should be interpreted as a literature-level reference rather than a strictly controlled head-to-head comparison. Nevertheless, the proposed method demonstrates a favorable balance between classification accuracy and model complexity.

4.5. Cross-validation for robustness evaluation

The 3-fold patient-wise cross-validation was conducted only as a post-hoc robustness analysis after the final architecture, preprocessing strategy, and hyperparameters had been fixed. It was not used for model selection, hyperparameter tuning, or modification of the proposed framework. To further validate the robustness of the proposed model with respect to different patient divisions, we conducted

Table 9. 3-fold patient-wise cross-validation accuracy.

Fold	Training patients	Validation patients	Validation accuracy
1	32	16	98.95%
2	32	16	99.15%
3	32	16	99.07%
Mean $\pm$ Std			99.06 $\pm$ 0.10%

3-fold patient-wise cross-validation. Each recording in the MIT-BIH database was regarded as an independent record-level subject, and all 48 subjects were randomly partitioned into three mutually exclusive folds, each containing 16 subjects. In each fold, two folds (32 subjects) were used as the training set, and the remaining fold (16 subjects) served as the validation set, thereby guaranteeing no patient overlap between the training and validation subsets. SMOTE oversampling was applied exclusively to the training set in each fold, whereas the validation set retained its original class distribution. Each fold was trained using an identical model architecture and fixed hyperparameters consistent with the main experiment. The results of the 3-fold cross-validation are presented in table 9.

The validation accuracies across the three folds range from 98.95% to 99.15%, with a mean accuracy of $99.06 \pm 0.10\%$. These results suggest that the proposed model is relatively insensitive to patient partition variations under the adopted experimental setting. However, this post-hoc cross-validation analysis was not used for model selection and should be regarded as supplementary evidence of robustness rather than definitive proof of the absence of overfitting.

4.6. Cross-dataset validation

To evaluate the portability and cross-dataset generalization ability of the proposed model, we selected the INCART Arrhythmia Database (St. Petersburg INCART 12-lead Arrhythmia Database) as an external validation set. This dataset consists of 75 12-lead ECG recordings with a sampling rate of 257 Hz, acquired at the Cardiology Research Institute in St. Petersburg, Russia. It exhibits substantial differences from the MIT-BIH database in terms of study population, recording equipment, and lead configuration, which makes it well-suited for validating cross-dataset generalization. For consistency with the MIT-BIH setting, the lead II signal of the INCART database was used as the input lead. Because the INCART recordings were sampled at 257 Hz, whereas the MIT-BIH recordings were sampled at 360 Hz, the heartbeat segmentation window for INCART was adjusted according to the sampling rate to maintain a time-equivalent segment length.

Experimental design:

Training: the CBiLSTM-ResNet model was trained only on the MIT-BIH training set (60% of patients under patient-wise splitting), without using any data from the INCART database.

Preprocessing: the INCART dataset was processed using the same preprocessing pipeline applied to the MIT-BIH dataset, including median filtering for baseline-drift suppression, 8-level db5 wavelet denoising, and heartbeat segmentation. For consistency with the main task setting of this study, only heartbeat categories mappable to the same four AAMI classes used in the MIT-BIH experiments, namely N, S, V, and F, were included in the external validation analysis.

For the INCART dataset, the identical AAMI beat mapping rules as applied to the MIT-BIH dataset (see table 1) were adopted. Specifically, the N class includes normal beats, left bundle branch block beats, right bundle branch block (RBBB) beats, atrial escape beats, and nodal escape beats (original symbols: N, L, R, e, j). The S class includes atrial premature beats, aberrated atrial premature beats, nodal premature beats, and supraventricular premature beats (original symbols: A, a, J, S). The V class includes ventricular premature beats and ventricular escape beats (original symbols: V, E). The F class includes fusion beats (original symbol: F). Beat types that could not be mapped to these four AAMI classes were excluded from the main classification analysis. The excluded symbols included Q (unclassifiable, $n = 6$), n (unclassifiable, $n = 32$), B (bundle branch block, $n = 1$), and + (rhythm change, $n = 12$), totaling 51 beats. Although the N class includes left and RBBB beats under the adopted AAMI mapping, the INCART symbol B represents an unspecified bundle branch block annotation and was not explicitly listed in the adopted four-class mapping table. Therefore, it was excluded from the main four-class evaluation. Because only one such beat was present, this exclusion had negligible influence on the external validation results. Notably, the inclusion criteria for the N class, including RBBB and supraventricular

Table 10. Beat counts included and excluded in the INCART dataset.

Category	Count
Included beats (mapped to AAMI classes)	
N	153,676
S	1960
V	20,013
F	219
Subtotal (included)	175,868
Excluded beats (unmappable)	
Q (unclassifiable)	6
n (unclassifiable)	32
B (bundle branch block)	1
+ (rhythm change)	12
Subtotal (excluded)	51
Total annotated beats	175,919

Table 11. Internal held-out and external cross-dataset validation results.

Test dataset	Overall accuracy	Precision (N)	Precision (S)	Precision (V)	Precision (F)	Recall (N)	Recall (S)	Recall (V)	Recall (F)
MIT-BIH	99.39%	99.87%	88.41%	95.91%	12.00%	99.56%	95.31%	97.50%	75.00%
INCART	98.75%	98.66%	94.49%	98.41%	80.00%	98.97%	96.05%	96.17%	85.96%

Table 12. Comparison with recent methods on the INCART database.

Method	INCART accuracy
KAN-based [28]	96.87%
xLSTM-based [28]	97.15%
MAK-Net [30]	99.48%
Proposed (CBiLSTM-ResNet, single-lead)	98.75%

escape beats, are strictly consistent with the MIT-BIH database, maintaining identical category definition standards across both datasets to ensure comparability.

For full transparency, the numbers of included and excluded beats are reported in table 10.

Testing: Preprocessed heartbeat samples from the INCART dataset were directly input into the pre-trained model without fine-tuning, domain adaptation, or any additional training.

Evaluation Metrics: Overall classification accuracy, class-wise precision, and class-wise recall.

The cross-dataset test results are summarized in table 11.

For the INCART external validation, only the 175 868 beats that could be mapped to the four AAMI classes, namely N, S, V, and F, were included in the calculation of accuracy, precision, and recall. The 51 unmapable annotated beats were reported for transparency but were excluded from the metric calculation, because they did not belong to the target four-class classification task.

The proposed model achieves an overall classification accuracy of 98.75% on the INCART database. Despite considerable discrepancies in recording devices, sampling frequency, and study population between the INCART and MIT-BIH databases, the model still achieved competitive cross-dataset performance, which suggests that the proposed model has promising cross-dataset generalization potential under heterogeneous recording conditions.

For reference, recent methods tested on the INCART database under similar MIT-BIH-trained settings reported accuracies of 96.87% (KAN), 97.15% (xLSTM), and 99.48% (MAK-Net), as summarized in table 12. Our single-lead model achieves 98.75%, which is competitive despite using less information than multi-lead approaches.

In terms of category-level performance, the model attains high recall values for normal beats (N) and ventricular ectopic beats (V), reaching 98.97% and 96.17% respectively, illustrating its reliable generalization ability for representative heartbeat categories. By comparison, the recall rates of supraventricular ectopic beats (S) and fusion beats (F) are relatively lower, measured at 96.05% and 85.96% accordingly. This phenomenon is mainly attributed to annotation inconsistencies and pronounced morphological

variations of these two categories across heterogeneous databases, which remains a common challenge in automatic ECG beat classification under heterogeneous database conditions.

In conclusion, the cross-dataset validation results suggest that the proposed model has favorable portability and generalization potential under heterogeneous recording conditions. However, broader validation on additional public and clinical ECG datasets is still needed before drawing stronger conclusions about its potential clinical applicability.

5. Conclusion

This study proposed an ECG beat classification framework that integrates a dilated multi-scale ResNet with BiLSTM-based temporal feature modeling. The multi-scale dilated convolutional module was designed to extract ECG morphological patterns at different temporal scales, while the BiLSTM layer was introduced to model temporal dependencies within the extracted heartbeat feature sequence. Through this design, the proposed framework jointly exploits local morphological information, multi-scale waveform representation, and sequential feature dependency for ECG beat classification.

Experimental results on the MIT-BIH Arrhythmia Database demonstrated that the proposed framework improves overall classification performance and enhances the recognition of minority and abnormal heartbeat categories, especially S and V beats. Because only four F beats were included in the held-out test set, the F-class results should be interpreted cautiously. The ablation studies verified the effectiveness of the multi-scale module, BiLSTM layer, training-set-only class balancing strategy, and dilated convolution. In addition, external validation on the INCART Arrhythmia Database suggested that the proposed model has promising cross-dataset generalization potential under heterogeneous recording conditions.

Nevertheless, several limitations should be acknowledged. First, although external validation was conducted on the INCART database, the current study was still limited to two public ECG datasets, and broader validation on larger multi-center clinical datasets is needed. Second, although the model learns features associated with ECG morphology and temporal feature dependency, these features remain latent representations rather than explicit clinical measurements. Third, although patient-wise splitting was adopted and the held-out test records were not used for model development, the preprocessing and model selection procedures were not evaluated using a fully nested cross-validation framework. Future work will focus on multi-center clinical validation, more interpretable feature analysis, and further optimization for real-time wearable or bedside ECG monitoring devices.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Availability of data and materials

The ECG recordings used in this study were obtained from the publicly available MIT-BIH Arrhythmia Database and the St. Petersburg INCART 12-lead Arrhythmia Database. Additional processed beat samples and implementation details are available from the corresponding author upon reasonable request.

Conflict of interest

The authors declare that they have no competing interests. The authors had full access to all data in the study and take full responsibility for the integrity of the data and the accuracy of the data analysis.

Ethics approval and consent to participate

This study used de-identified ECG recordings from the publicly available MIT-BIH Arrhythmia Database and the St. Petersburg INCART 12-lead Arrhythmia Database. According to institutional and national guidelines, secondary analysis of anonymized open-access data does not require additional approval from an ethics committee, and the requirement for informed consent was therefore waived.

Authors' contributions

S Q and L H conceived and designed the study. S Q implemented the ECG preprocessing pipeline, constructed the dilated multi-scale ResNet-BiLSTM model, and performed all experiments. L H contributed to network architecture design, parameter optimization, and interpretation of the experimental results. X Z curated the MIT-BIH arrhythmia data, verified the annotation-to-AAMI class mapping, and conducted data cleaning and consistency checking. Y Z assisted with model training, statistical analysis, and visualization of the results. N L supervised the overall research process, provided critical revision of the manuscript, and coordinated project administration. All authors contributed to manuscript editing, read and approved the final version, and agreed to be accountable for all aspects of the work.

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